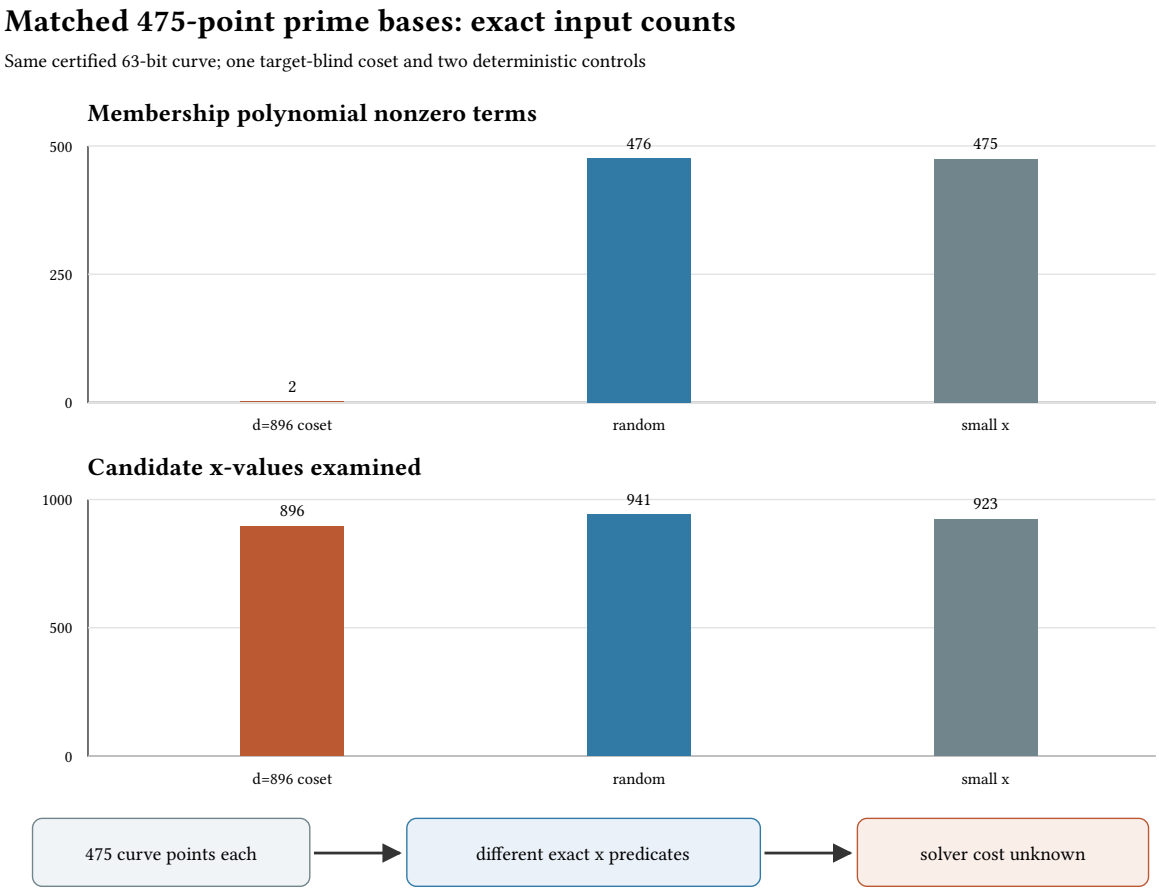
Equal-realized-size controls for a 475-point prime coset base

Exact target-blind preparation and independent Sage replay, 2026-10-17 UTC. Two deterministic controls each contain 475 curve points on the certified 63-bit prime-order curve. The control record keeps every random draw and small-x candidate, including misses, and the full membership coefficients. No solver degree, relation yield or ECDLP speed was measured.



Control record SHA-256: 8b52f3e7878357d83c80996780d904f8f88f8ffd4e0b69d78597c5c3bce2a68a

The d=896 coset predicate has two terms but 896 field roots, of which 421 do not lift to curve points. The random control's exact degree-475 product has 476 nonzero terms; the small-x product has 475 because x=0 makes its constant coefficient zero. The random control examined 941 x-values and small-x examined 923. These inputs are matched in realized point count and curve, not in algebraic representation or measured runtime.

Independent check and failure custody

SageMath 10.10.rc0 replayed both draw streams, every accepted curve point, the point digests and every membership coefficient. The first builder attempt failed an overstrong 476-term assertion; the first Sage replay failed only when writing a Sage integer to JSON. The partial output is retained, and the corrected replay is a new file. Neither failure supplies negative evidence about the factor-base mechanism.

Required next comparison

Freeze control systems on a common target schedule and independently check point extraction. A bounded solver protocol must then measure actual degree, time and RSS, certified natural relations, useful rank, linear algebra, unseen same-point target cost and matched rho, charging all setup and the coset's nonliftable roots. The companion Markdown report links the full records and failed-attempt artifact.